

Solved Problem on a one-dimensional quantum harmonic oscillator

Problem #1

Problem Statement

A one-dimensional quantum harmonic oscillator at time $t = 0$ is in the state

$$|\psi_\alpha\rangle = \cos \alpha |0\rangle + e^{i\phi} \sin \alpha |1\rangle.$$

Answer the following:

- (a) Calculate the expectation values

$$\langle x \rangle(t), \quad \langle p \rangle(t), \quad \langle E \rangle.$$

- (b) Calculate the uncertainties

$$\Delta x(t), \quad \Delta p(t), \quad \Delta E.$$

Solution

Throughout the solution we use the harmonic oscillator Hamiltonian

$$\hat{H} = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right)$$

whose energy eigenstates satisfy

$$\hat{H}|n\rangle = \hbar\omega \left(n + \frac{1}{2} \right) |n\rangle.$$

The ladder operators obey

$$a|n\rangle = \sqrt{n} |n-1\rangle,$$

$$a^\dagger|n\rangle = \sqrt{n+1} |n+1\rangle.$$

The position and momentum operators are

$$x = \sqrt{\frac{\hbar}{2m\omega}}(a + a^\dagger)$$

and

$$p = -i\sqrt{\frac{m\hbar\omega}{2}}(a - a^\dagger).$$

Step 1: Time evolution of the state

Since each energy eigenstate acquires only a phase,

$$|n, t\rangle = e^{-iE_n t/\hbar} |n\rangle,$$

where

$$E_0 = \frac{1}{2}\hbar\omega, \quad E_1 = \frac{3}{2}\hbar\omega.$$

Therefore

$$|\psi(t)\rangle = \cos \alpha e^{-i\omega t/2} |0\rangle + e^{i\phi} \sin \alpha e^{-i3\omega t/2} |1\rangle.$$

Factoring out the global phase,

$$|\psi(t)\rangle = e^{-i\omega t/2} \left[\cos \alpha |0\rangle + e^{i(\phi - \omega t)} \sin \alpha |1\rangle \right].$$

Since a global phase has no physical effect, only the relative phase

$$\phi - \omega t$$

appears in expectation values.

Part (a)

Expectation value of position

Write

$$x = x_0(a + a^\dagger),$$

where

$$x_0 = \sqrt{\frac{\hbar}{2m\omega}}.$$

First compute

$$\langle a \rangle.$$

The only non-zero contribution is

$$\langle 0|a|1\rangle = 1.$$

Hence

$$\langle a \rangle = \cos \alpha e^{-i(\phi - \omega t)} \sin \alpha.$$

Similarly,

$$\langle a^\dagger \rangle = \cos \alpha e^{i(\phi - \omega t)} \sin \alpha.$$

Therefore

$$\begin{aligned} \langle x \rangle &= x_0(\langle a \rangle + \langle a^\dagger \rangle) \\ &= 2x_0 \cos \alpha \sin \alpha \cos(\phi - \omega t). \end{aligned}$$

Using

$$2 \cos \alpha \sin \alpha = \sin 2\alpha,$$

we obtain

$$\langle x \rangle(t) = \sqrt{\frac{\hbar}{2m\omega}} \sin 2\alpha \cos(\phi - \omega t).$$

Expectation value of momentum

Using

$$p = -ip_0(a - a^\dagger),$$

where

$$p_0 = \sqrt{\frac{m\hbar\omega}{2}},$$

we obtain

$$\begin{aligned} \langle p \rangle &= -ip_0(\langle a \rangle - \langle a^\dagger \rangle) \\ &= -2p_0 \cos \alpha \sin \alpha \sin(\phi - \omega t). \end{aligned}$$

Hence

$$\langle p \rangle(t) = -\sqrt{\frac{m\hbar\omega}{2}} \sin 2\alpha \sin(\phi - \omega t).$$

Expectation value of energy

Since the Hamiltonian is diagonal in the number basis,

$$\begin{aligned} \langle E \rangle &= \cos^2 \alpha \left(\frac{1}{2} \hbar\omega \right) + \sin^2 \alpha \left(\frac{3}{2} \hbar\omega \right) \\ &= \frac{\hbar\omega}{2} (1 + 2 \sin^2 \alpha). \end{aligned}$$

Therefore

$$\langle E \rangle = \frac{\hbar\omega}{2} (1 + 2 \sin^2 \alpha).$$

Notice that the expectation value of energy is constant in time, as expected.

Part (b)

Position uncertainty

We first calculate

$$x^2 = x_0^2(a + a^\dagger)^2.$$

Expanding,

$$(a + a^\dagger)^2 = a^2 + a^{\dagger 2} + aa^\dagger + a^\dagger a.$$

Since

$$\langle a^2 \rangle = \langle a^{\dagger 2} \rangle = 0,$$

only the last two terms contribute.

Using

$$aa^{\dagger} = N + 1, \quad a^{\dagger}a = N,$$

we obtain

$$\langle x^2 \rangle = x_0^2(2\langle N \rangle + 1).$$

Now

$$\langle N \rangle = \sin^2 \alpha.$$

Hence

$$\langle x^2 \rangle = \frac{\hbar}{2m\omega}(1 + 2\sin^2 \alpha).$$

Therefore

$$(\Delta x)^2 = \langle x^2 \rangle - \langle x \rangle^2.$$

Substituting,

$$(\Delta x)^2 = \frac{\hbar}{2m\omega} [1 + 2\sin^2 \alpha - \sin^2 2\alpha \cos^2(\phi - \omega t)].$$

Thus

$$\Delta x = \sqrt{\frac{\hbar}{2m\omega} [1 + 2\sin^2 \alpha - \sin^2 2\alpha \cos^2(\phi - \omega t)]}.$$

Momentum uncertainty

Similarly,

$$p^2 = -p_0^2(a - a^{\dagger})^2.$$

One finds

$$\langle p^2 \rangle = \frac{m\hbar\omega}{2}(1 + 2\sin^2 \alpha).$$

Hence

$$(\Delta p)^2 = \langle p^2 \rangle - \langle p \rangle^2.$$

Therefore

$$(\Delta p)^2 = \frac{m\hbar\omega}{2} [1 + 2\sin^2 \alpha - \sin^2 2\alpha \sin^2(\phi - \omega t)].$$

or

$$\Delta p = \sqrt{\frac{m\hbar\omega}{2} [1 + 2\sin^2 \alpha - \sin^2 2\alpha \sin^2(\phi - \omega t)]}.$$

Energy uncertainty

Since

$$H^2|n\rangle = E_n^2|n\rangle,$$

we have

$$\begin{aligned}\langle H^2 \rangle &= \cos^2 \alpha \left(\frac{1}{2} \hbar \omega \right)^2 + \sin^2 \alpha \left(\frac{3}{2} \hbar \omega \right)^2 \\ &= (\hbar \omega)^2 \left(\frac{1}{4} + 2 \sin^2 \alpha \right).\end{aligned}$$

Now

$$(\Delta E)^2 = \langle H^2 \rangle - \langle H \rangle^2.$$

Substituting the expectation value,

$$\boxed{(\Delta E)^2 = (\hbar \omega)^2 \sin^2 \alpha \cos^2 \alpha.}$$

Using

$$\sin 2\alpha = 2 \sin \alpha \cos \alpha,$$

we finally obtain

$$\boxed{\Delta E = \frac{\hbar \omega}{2} \sin 2\alpha.}$$

Since an uncertainty is always non-negative, one may also write

$$\boxed{\Delta E = \frac{\hbar \omega}{2} |\sin 2\alpha| .}$$

Final Answers

$$\boxed{\langle x \rangle(t) = \sqrt{\frac{\hbar}{2m\omega}} \sin 2\alpha \cos(\phi - \omega t)}$$

$$\boxed{\langle p \rangle(t) = -\sqrt{\frac{m\hbar\omega}{2}} \sin 2\alpha \sin(\phi - \omega t)}$$

$$\boxed{\langle E \rangle = \frac{\hbar \omega}{2} (1 + 2 \sin^2 \alpha)}$$

$$\boxed{\Delta x = \sqrt{\frac{\hbar}{2m\omega}} [1 + 2 \sin^2 \alpha - \sin^2 2\alpha \cos^2(\phi - \omega t)]}$$

$$\boxed{\Delta p = \sqrt{\frac{m\hbar\omega}{2}} [1 + 2 \sin^2 \alpha - \sin^2 2\alpha \sin^2(\phi - \omega t)]}$$

$$\boxed{\Delta E = \frac{\hbar \omega}{2} |\sin 2\alpha|}$$